

INFO 510 Bayesian Modelling and Inference

Lecture 11 – Markov chains

(MCMC algorithm)

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Markov Chain Monte Carlo

- MCMC methods are a family of algorithms that uses Markov Chains to perform Monte Carlo estimate
- Class of algorithms used to draw samples from a probability distribution
- Important and popular concepts in Bayesian Statistics, especially while doing inference

Markov chains



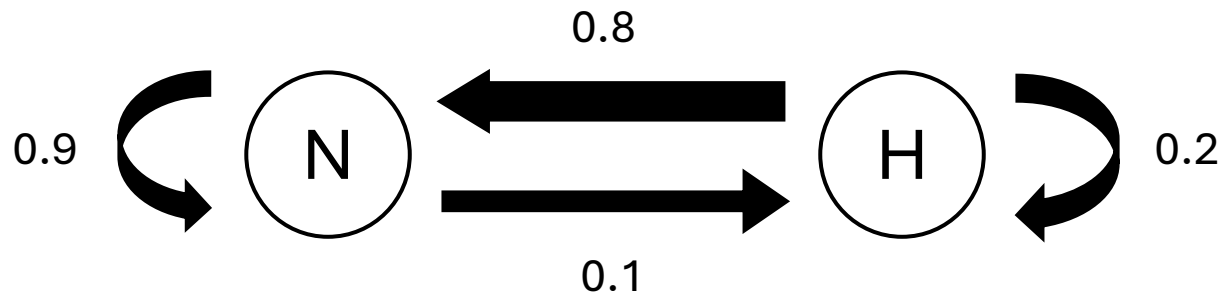
Some chocolate

Nestle = 0.6

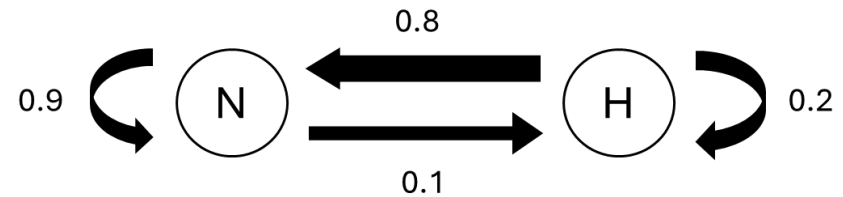


Sum chocolate

Hershey = 0.4



Nestle = 0.6
Hershey = 0.4



Market share today (time t)

given the next unit of time ($t+1$),

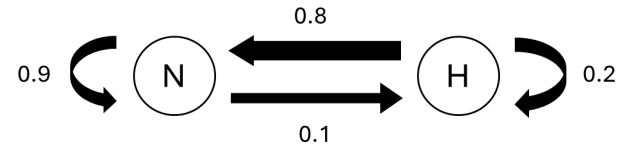
0.9 or 90% of Nestle customers stay back with Nestle

0.2 or 20% of Hershey customers stay back with Hershey

0.1 or 10% of Nestle customers are switching to Hershey

0.8 or 80% of Hershey customers are switching to Nestle

Transition of states



Nestle = 0.6

Hershey = 0.4

		t + 1	
		N	H
t	N	0.9	0.1
	H	0.8	0.2

Transition matrix

At time t + 1, what market share will Nestle and Hershey have?

$$\begin{aligned}
 N(t + 1) &= (0.6 * 0.9) + (0.4 * 0.8) \\
 &= 0.54 + 0.32 = 0.86
 \end{aligned}$$

$$\begin{aligned}
 H(t + 1) &= (0.4 * 0.2) + (0.6 * 0.1) \\
 &= 0.08 + 0.06 = 0.14
 \end{aligned}$$

	N	H
t	0.6	0.4
t + 1		
t + 2		

Markov chain is a process that consists of a finite number of states and some known probabilities p_{ij} , where p_{ij} is the probability of moving from state i to state j . Transition probability (transition matrix)

Markov Property and Memorylessness

The key feature of Markov Chains is that the future state depends only on the current state, not on the sequence of events that preceded it. This property is called the Markov Property.

$$P(X_{t+1} = s \mid X_t = s_t, \cancel{X_{t-1} = s_{t-1}}, \dots, \cancel{X_0 = s_0}) = P(X_{t+1} = s \mid X_t = s_t)$$

distribution
of X_{t+1}

depends
on X_t

whatever happened before
time t doesn't matter.

Transition Matrix

The transition matrix is like a map of all possible moves and their probabilities

A transition matrix is a table (or matrix) that tells us the probability of moving from one state to another in a single step.

Each entry in this matrix represents the probability of moving from one state to another.

if you have **n states**, the transition matrix **P** is an **n × n** matrix. Each element in the matrix, **P_{ij}**, represents the probability of moving from state **i** to state **j**

Each **row** corresponds to the current state

Each **column** corresponds to the next state

Values in the matrix are probabilities, so they are numbers between 0 and 1

$$P = \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{bmatrix}$$

Example – Transition matrix

Let's assume we have three states, S_1 , S_2 , and S_3

S_1 = Sunny; S_2 = Cloudy; S_3 = Rainy

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.4 & 0.4 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{bmatrix} \quad \text{each row adds up to 1 (or 100\%)}$$

$P_{12} = 0.3$: This means that if it's sunny today (S_1), there's a 30% chance it will be cloudy (S_2) tomorrow.

$P_{33} = 0.3$: If it's rainy today (S_3), there's a 30% chance it will still be rainy tomorrow.

Matrix power, Long-Term Predictions

By multiplying the transition matrix by itself (called matrix powers), you can predict the probability of the system being in each state multiple steps into the future

if you want to know the state probabilities after k steps, you raise the transition matrix P to the power of k : P^k

P^2 : Tells you the probability of being in any state after two transitions

P^3 : Tells you the probabilities after three transitions, and so on

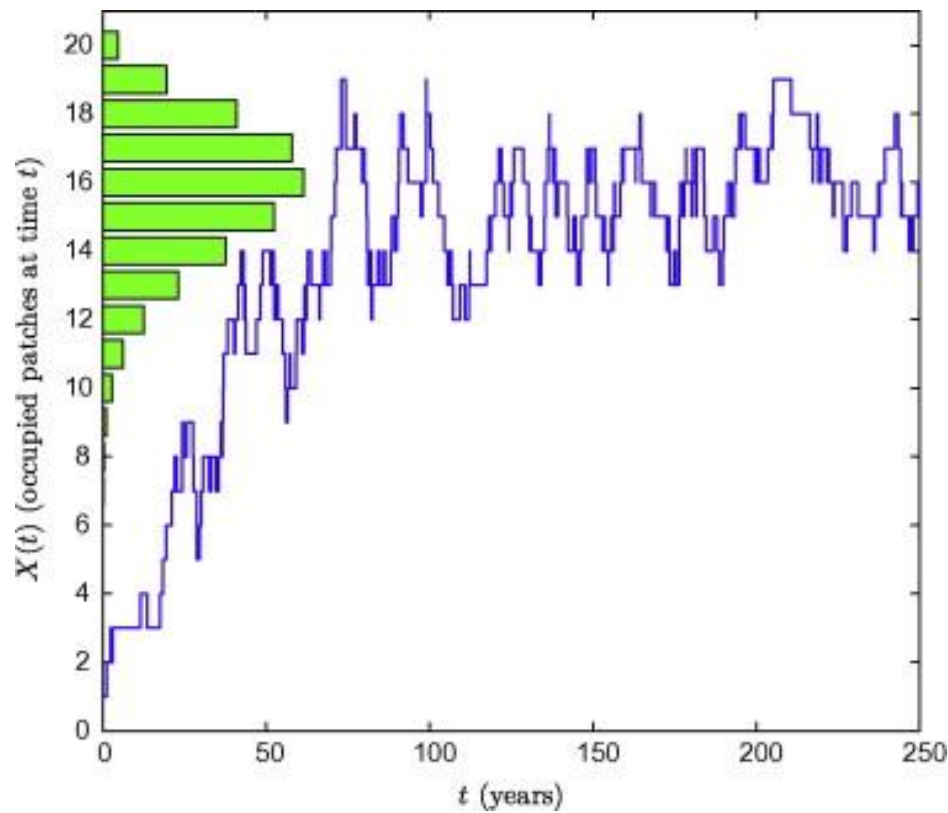
Allows us to make long term predictions

Stationary Distribution - Where the System Settles

After many steps, a Markov chain may reach a state where the probabilities stop changing

Stationary Distribution - Where the System Settles

After many steps, a Markov chain may reach a state where the probabilities stop changing



Properties of Markov Chain or classification of states

A state in a Markov chain is called **Absorbing** if there is no possible way to leave that state. Absorbing states do not have any outgoing transitions from it.

A state in a Markov chain is said to be **Transient** if there is a non-zero probability that the chain will never return to the same state, otherwise, it is **Recurrent**.

A state in a Markov chain is said to be **Periodic** if returning to it requires a multiple of some integer larger than 1, the greatest common divisor of all the possible return path lengths will be the period of that state.

A Markov chain is said to be **Irreducible** if we can go from one state to another in a single or more than one step.

Absorbing States

A state is called *absorbing* if, once entered, it cannot be left. In other words, the probability of leaving an absorbing state is zero.

$$P[i,i]=1, \text{ and for all other states } j, P[i,j]=0$$

In a transition matrix, an absorbing state will have a 1 in its diagonal (meaning a 100% chance of staying in that state) and 0s in the rest of the row (meaning no chance of moving to other states).

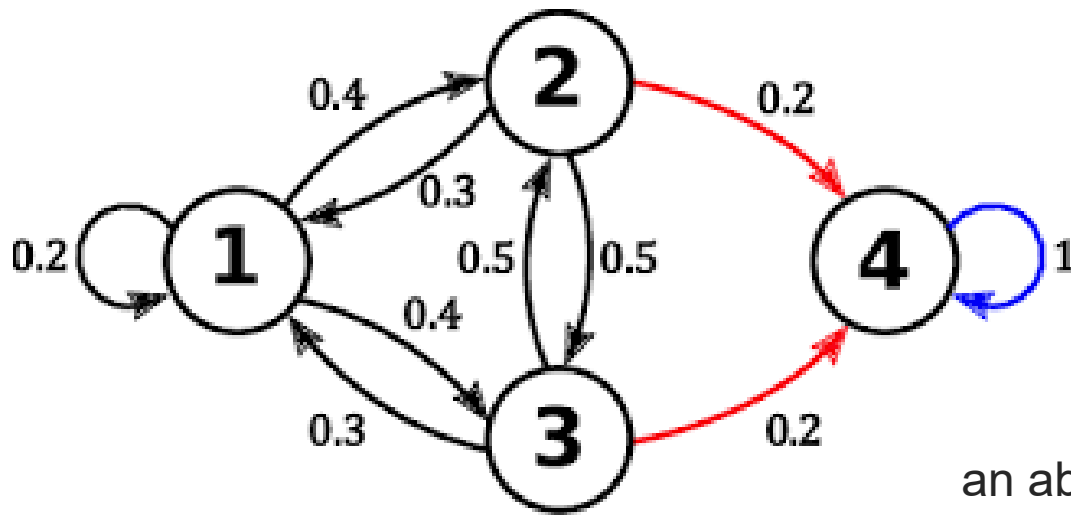
State 1: Working

State 2: Unemployed

State 3: Retired (absorbing state)

State 4: Bankrupt (another absorbing state)

$$P = \begin{bmatrix} 0.7 & 0.3 & 0 & 0 \\ 0.4 & 0.3 & 0.2 & 0.1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



an absorbing Markov chain with 4 states,
where S4 is an absorbing state

Applications of Absorbing Markov chains

Health: A medical condition with different stages. For example, in the progression of a disease, "death" is an absorbing state

Economics: Bankruptcy is an absorbing state for companies. Once a company is bankrupt, it's out of the market

Customer Service: In customer service, a "resolved" state can be an absorbing state. Once a customer issue is marked as resolved, that's the end of the process

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Recurrent and Transient states

A recurrent state is a state that you always come back to if you enter it. Once you reach it, you have a guaranteed chance of returning eventually.

A transient state is a state that you might leave and never come back to. You might pass through it, but there's a chance you never revisit it.

Guaranteed return - If you're in a recurrent state, you're certain to return to that state at some point in the future. You might take a detour through other states, but you will come back

Infinite visits - You'll visit a recurrent state an infinite number of times over a long enough period of time

Not guaranteed to return - If you're in a transient state, there's a chance you'll never come back once you leave

Finite visits - You might visit a transient state a few times, but after a long enough period, you'll eventually stop returning to it

Recurrence Test:

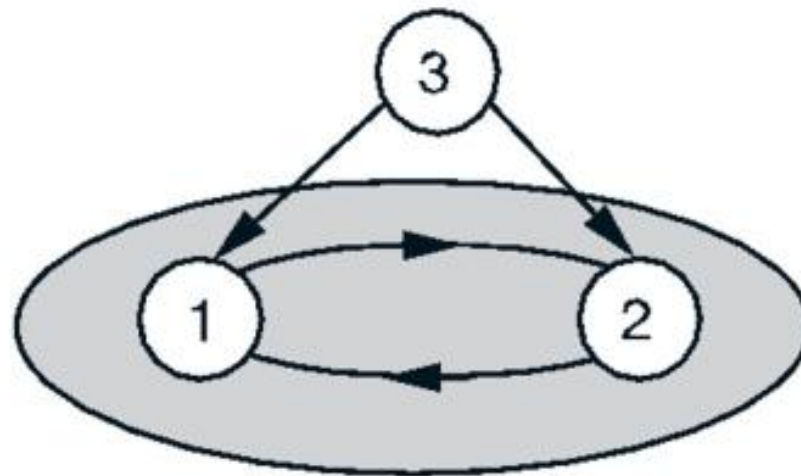
- If the expected number of returns to a state is infinite, the state is recurrent.
- If you sum the probabilities of returning to state i over all time steps, and the sum is infinite, then the state is recurrent.

$$\sum_{n=1}^{\infty} P(\text{return to state } i \text{ after } n \text{ steps}) = \infty$$

Transience Test:

- If the expected number of returns is finite, the state is transient.
- If you sum the probabilities of returning to state i over all time steps and the sum is finite, then the state is transient.

$$\sum_{n=1}^{\infty} P(\text{return to state } i \text{ after } n \text{ steps}) < \infty$$



Single class of recurrent states (1 and 2)
and one transient state (3)

3 states for status of a company at the stock market

Stable, At Risk, and Bankrupt.

If the company can always recover and return to being Stable, then Stable is a recurrent state

If the company goes Bankrupt, and once it's bankrupt, it's out of the game, then Bankrupt is an absorbing state (since once you enter, you never leave).

If the company is At Risk and it can either stabilize or go bankrupt, but there's a chance it won't return to At Risk, then At Risk is a transient state

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Periodic state

A state has period d if it can only be revisited after a fixed number of steps that is a multiple of d

once you leave a periodic state, you can only return to it after a certain number of steps, and this number is the same every time.

$$d(x) = \gcd\{n \in \mathbb{N} : P^n(x, x) > 0\}$$

If $d(x)=1$, the state is aperiodic, meaning you can return to the state at irregular intervals, so you don't have to wait for a fixed number of steps

If $d(x)>1$, the state is periodic with period d . This means that once you leave the state, you will only return to it after a number of steps that is a multiple of d

E.g.,

- your home is state i , and the period of your home is 3 (i.e., $d(i)=3$)
- You leave your home and start walking. After 1 or 2 steps, you can't return home - you must wait until exactly 3 steps to return home
- Then, after leaving again, you'll always return after 3 steps, 6 steps, 9 steps, and so on. These return times are multiples of 3, making the state periodic with period 3

For state i with period $d(i)=3$, the probabilities $P_{ii}(n)$ (the probability of returning to state i after n steps) will look like this:

- $P_{ii}(1)=0$ (can't return after 1 step),
- $P_{ii}(2)=0$ (can't return after 2 steps),
- $P_{ii}(3)>0$ (there's a chance you return after 3 steps),
- $P_{ii}(6)>0$ (there's a chance you return after 6 steps),
- $P_{ii}(9)>0$, and so on.

Why Periodicity Matters in Markov Chains

Periodicity affects the long-term behavior of the Markov chain

For example, in an aperiodic Markov chain, the system tends to "forget" its starting point after a long time, which leads to the possibility of reaching a steady-state distribution (a state where the probabilities of being in each state stabilize over time)

However, if a Markov chain is periodic, then the chain doesn't have a steady state in the usual sense because the system keeps revisiting certain states in cycles

This periodicity can prevent the system from converging to a single distribution over the states

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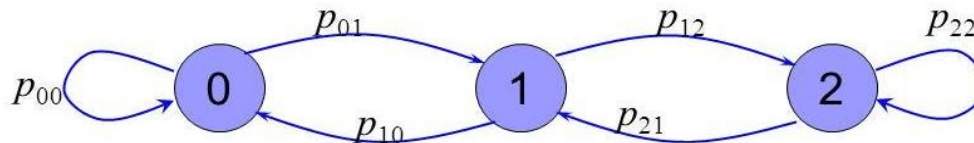
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Irreducible states

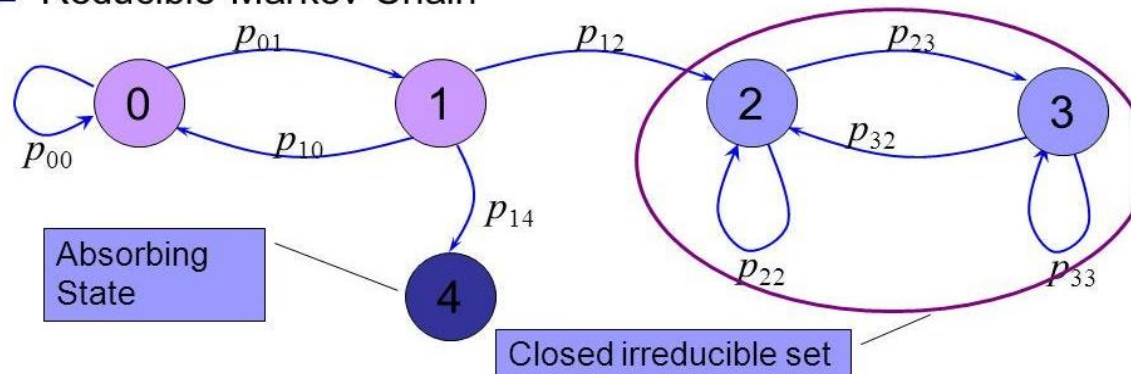
It's possible to get from any state to any other state, either directly or through a series of intermediate states

the system is fully "connected" in the sense that no state is isolated — given enough steps, you can travel between any two states

■ Irreducible Markov Chain



■ Reducible Markov Chain



A Markov chain is **irreducible** if, for any two states i and j , there exists an integer $n \geq 1$ such that

$$P_{ij}^{(n)} > 0$$

$P_{ij}^{(n)}$ is the probability of moving from state i to state j in exactly n steps

This means there is a nonzero probability of getting from state i to state j in some finite number of steps.

The term "irreducible" comes from the idea that the Markov chain cannot be broken down into smaller, disconnected parts.

Properties of Irreducible Markov Chains

1. Every State Is Recurrent or Transient Together

If one state is recurrent (meaning you'll always return to it), then all states in the chain are recurrent

If one state is transient (meaning you might leave and never return), then all states are transient

2. Unique Stationary Distribution (if irreducible and aperiodic)

This means that as time goes to infinity, the probabilities of being in each state stabilize, and the system reaches a long-term equilibrium

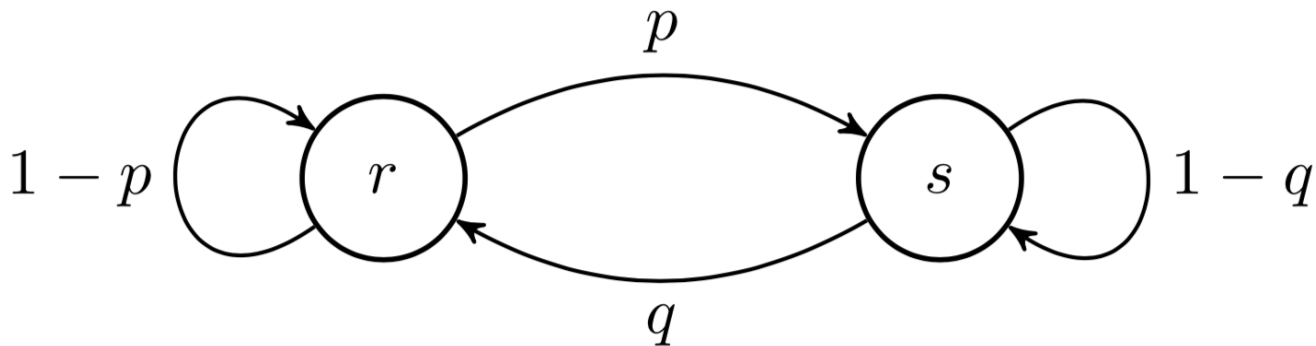
3. Positive Recurrence

the states are positive recurrent - means that you'll return to each state within a finite expected time

Ergodic Markov Chains

A Markov Chain is *ergodic* if it is both irreducible (all states can communicate) and aperiodic (there is no fixed period for returning to a state)

If a Markov chain is ergodic, it has a unique stationary distribution, meaning that, as time goes on, the chain settles into a long-term stable behavior where the probability of being in each state stabilizes



A Markov chain is ergodic if it satisfies these two key properties:

Irreducibility

Aperiodicity

Long-term stable probabilities -

Over time, the probability of being in each state reaches a stable value, regardless of the initial state. These probabilities form the stationary distribution of the chain.

Positive recurrence means that you expect to return to a particular state within a finite number of steps.

Example:

A weather system where the states represent "sunny," "rainy," and "cloudy" days. If the system is ergodic, over time the probability of a sunny day, a rainy day, or a cloudy day stabilizes, and no matter what the weather is today, the long-term distribution of weather states is predictable.

First-Step Analysis and Hitting Times

First-Step Analysis: This technique involves breaking down a complex Markov Chain problem into simpler "first-step" decisions. Instead of analyzing the entire chain at once, you compute the expected values based on the first step and then solve the recursive equation.

Using First Step Analysis, you calculate the overall time by averaging over all the possible routes, weighted by how likely you are to take each first step

Hitting Times

The hitting time (also called the first-passage time) is the number of steps it takes for a Markov chain to reach a particular state for the first time, starting from a different state

Expected hitting time: How long, on average, will it take to reach a specific state from a starting state?

Hitting probability: What is the probability of eventually reaching a specific state from a starting state?

Let's define T_i as the expected time to reach state k starting from state i . The first step analysis approach gives the equation:

$$T_i = 1 + \sum_k P_{ik} T_k$$

- 1 is the time of the **first step**.
- P_{ik} is the probability of transitioning from state i to state k in one step
- T_k is the expected hitting time starting from state k

If we want to compute the probability of reaching a target state k from state i , call this probability h_i , First Step Analysis gives:

$$h_i = \sum_k P_{ik} h_k$$

- P_{ik} is the probability of transitioning from i to k
 h_k is the probability of reaching j starting from k

Jumping between 3 states – 0, 1, and 2

State 0 is absorbing - Once you reach state 0, stay there forever

From state 1

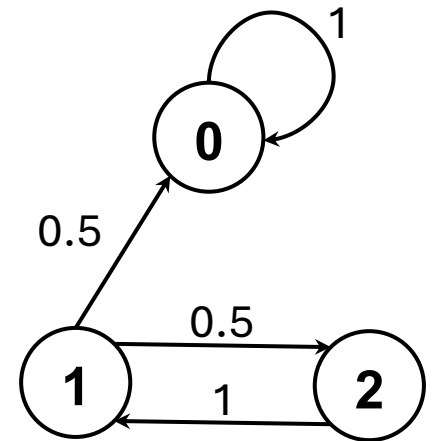
- You move to state 0 with probability 0.5 (and is absorbed)
- You move to state 2 with probability 0.5

From state 2 - you move to state 1 with probability 1

The process continues until you reach state 0

We want to compute:

1. Expected hitting time h_1 : How long does it take to reach state 0, starting from state 1?
2. Expected hitting time h_2 : How long does it take to reach state 0, starting from state 2?



Define the Hitting Time Equations

h_1 be the expected number of jumps to reach state 0 from state 1

h_2 be the expected number of jumps to reach state 0 from state 2

set up equations using first-step analysis

$h_0 = 0$ (since state 0 is absorbing, it takes 0 steps to reach itself)

$$h_2 = 1 + h_1;$$

$$h_1 = 1 + 0.5h_0 + 0.5h_2$$

$$h_1 = 1 + 0.5(0) + 0.5h_2$$

$$h_1 = 1 + 0.5h_2$$

$$h_1 = 1 + 0.5(1 + h_1)$$

$$h_1 = 1 + 0.5 + 0.5h_1$$

$$h_1 = 3$$

Now, solving for h_2

$$h_2 = 1 + h_1$$

$$h_2 = 1 + 3$$

$$h_2 = 4$$

- The expected number of steps to reach state 0 from state 1 is **3**
- The expected number of steps to reach state 0 from state 2 is **4**

- The minimum possible steps to reach state 0 is 1 step. The expected hitting time is 3 steps, because sometimes it gets "stuck" at state 2 before reaching 0
- Hitting time is NOT the minimum steps - it's the expected number of steps needed, considering the stochastic nature of the Markov chain.
- If there's randomness, the path taken varies, and hitting time accounts for the average behavior, not just the shortest path.
- Hitting time is useful for predicting system behavior when a process is probabilistic rather than deterministic.
- E.g., How long before an infectious disease spreads to a new population.

Applications of Markov Chains

Queueing Systems

Markov Chains are used to model waiting lines, where the next customer's arrival or the next service completion depends only on the current state of the queue

Finance

Credit ratings of companies can be modeled as a Markov Chain, where the company's future rating depends only on its current rating

Genetics

Markov Chains are used to model gene sequences, where the likelihood of a particular nucleotide appearing at a position depends on the preceding nucleotide.

Markov Models



	State is "Fully Observable" (or: "Fully Simulated")	State is Partially Observable
System is Autonomous	Markov Chain	Hidden Markov Model
System is Controlled	Markov Decision Process	Partial Observable MDP